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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO5-AT2-TECHNICAL PRESENTATION

Algorithm & Logic Identification

COURSE CODE: CSA1709

Guided By:
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Presented by:
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SIMATS ENGINEERING

CO5- ASSESSMENT TOOL 2 – Technical Presentation

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Code of Conduct:

I P. Ramesh (192312443) that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *P. Ramesh*

1. **System Analysis Presentation**

   Present a reverse engineering analysis of an AI-based system, explaining its architecture, components, and working process using diagrams.

2. **Architecture Reconstruction Demo**

   Prepare a technical presentation to reconstruct the architecture of a given software/AI system, highlighting module interactions and data flow.

3. **Algorithm & Logic Identification**

   Present how you identified algorithms and decision-making logic in a reverse-engineered system, including methodology and tools used.

4. **Data Flow & Processing Pipeline**

   Explain through a presentation the data flow, processing pipeline, and communication mechanisms in a reverse-engineered distributed system.

5. **Enhancement & Redesign Proposal**

   Deliver a technical presentation proposing improvements or redesign strategies for an existing system based on reverse engineering findings.

RUBRICS FOR CALCULATION

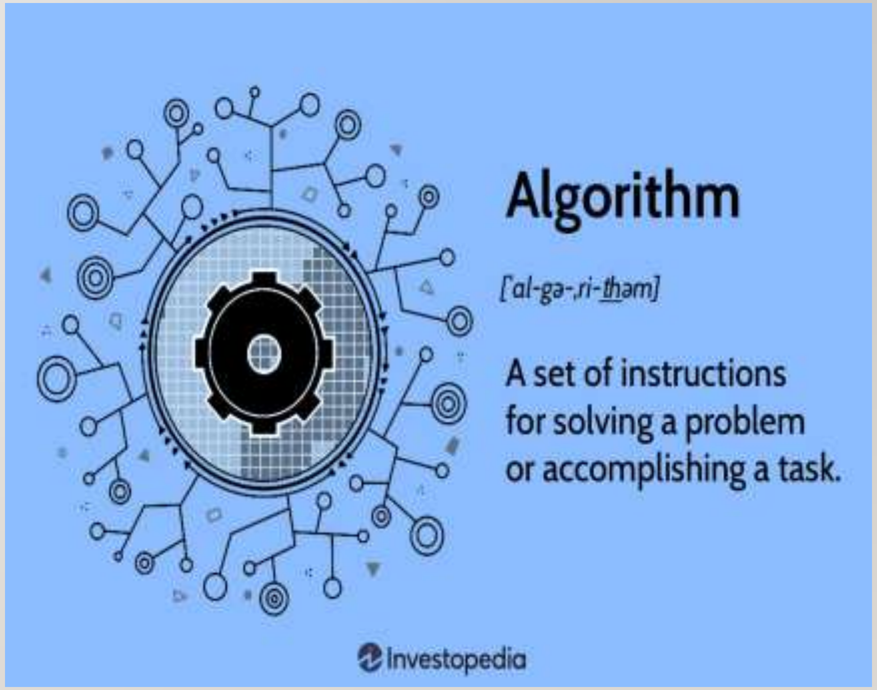
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

INTRODUCTION

- Algorithm & Logic Identification is the process of understanding how a reverse-engineered system processes data and makes decisions.
- It involves studying the system's inputs, outputs, processing steps, and decision rules.
- Different algorithms are identified by analyzing the system's behavior and computational patterns.
- Decision-making logic is examined using conditions, rules, thresholds, and state transitions.
- Tools such as Python, PySpark, flowcharts, pseudocode, and data-analysis techniques help reconstruct the system logic.

SYSTEM UNDERSTANDING

- Analyze the existing system architecture and workflow.
- Identify the main modules and their functions.
- Determine the input data required by the system.
- Study how input data is processed and transformed.
- Identify the outputs and expected results.
- Trace the data flow between different modules.



ALGORITHM IDENTIFICATION

- Analyze the input and output behavior of the existing system.
- Study the sequence of processing steps and calculations.
- Identify repeated patterns, loops, and computational operations.
- Examine conditions and rules used during processing.
- Compare the observed behavior with known algorithms.
- Identify algorithms related to classification, optimization, search, prediction, or decision-making.

DECISION-MAKING LOGIC

- Identify the rules and conditions used by the system to make decisions.
- Analyze if-else statements, thresholds, and decision criteria.
- Determine how the system selects an action from multiple alternatives.
- Study how different inputs lead to different decisions.
- Identify priority rules and constraints applied during decision-making.

Basic Decision Flow:

Input Data
↓
Data Processing
↓
Check Conditions
↓
Apply Decision Rules
↓
Select Best Action
↓
Generate Output

ALGORITHM RECONSTRUCTION

- Convert the identified system behavior into a clear algorithm or pseudocode.
- Define the required inputs, outputs, variables, and processing steps.
- Recreate the identified decision-making rules and conditions.
- Develop a flowchart to represent the complete algorithm.
- Implement the reconstructed algorithm using Python or PySpark.

RESULTS & VALIDATION

- Test the reconstructed algorithm using different input datasets and test cases.
- Compare the original system output with the reconstructed system output.
- Measure performance using accuracy, error rate, execution time, or optimization score.
- Check whether the identified decision-making logic produces the expected results.
- Identify differences, errors, or unexpected outputs.
- Modify and refine the algorithm based on the test results.
- Perform repeated tests to ensure consistent and reliable performance.
- Validate the final system under different operating conditions.

CONCLUSION

- Algorithm identification determines how the system processes information.
- Decision-logic analysis explains how the system makes decisions.
- Input-output analysis helps understand the system's internal behavior.
- System observation helps identify algorithms, rules, and processing steps.
- Python and PySpark support algorithm implementation and data processing.
- Flowcharts and pseudocode help represent and reconstruct system logic.

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